

Sewon Kim

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OBJECTIVE

Optimal Control and Robot Learning. My goal is to build a broad-sense control framework for robotic systems that exceeds biological behavior, inspired and enhanced by model-based and learning-based methods.

EDUCATION

University of Seoul

B.E. in Electrical and Computer Engineering, Minor in Mathematics

Mar 2022 – Aug 2027 (Expected)

Seoul, South Korea

EXPERIENCE

Drone Researcher at Institute of Innovation for Future Army, ROKA

Compulsory Military Service

Jan 2025 – Jul 2026

Daejeon, South Korea

AI Lead Engineer at InBody AI Scale Team, InBody

Jan 2024 – Jan 2025

Seoul, South Korea

Fifteenth Lead Researcher at Control and Dynamic Systems Lab, UOS

Advisor: Prof. Gyunghoon Park

Jan 2023 – Dec 2024

Seoul, South Korea

PROJECTS

- **Robot manipulation automating cable assembly and wiring**
AI for Industry Challenge
Nov 2025 – Present
Mountain View, USA
- **Autonomous air purifying robot agent**
4th Place (200 teams) — SK NAMUH X A1: Autonomous Driving Path Optimization (\$2,200)
Independently implemented full-stack autonomous navigation purely in NumPy, entirely without open-source code, covering MCL, MPPI, and Traveling Salesman Problem.
Aug 2025 – Sep 2026
Seoul, South Korea
- **Autonomous racing "Fifteenth"**
5th Place (12 teams) — 22nd Fifteenth Autonomous Grand Prix at CDC 2024
Led the project with 2 undergrads and implemented a full-stack physical autonomous navigation system for racing completely from scratch, covering SLAM, MCL, Trajectory Opt, MPPI.
Mar 2024 – Dec 2024
Milan, Italy
- **Cart pole system design and control by PID and NMPC**
3rd Place (25 teams) — UOS ECE Innovation Fair (\$220)
Independently developed a physical Cart-Pole system for \$272, covering modeling, motor control, sensor integration, state estimation, embedded systems, and MPC.
Jul 2024 – Sep 2024
Seoul, South Korea
- **New fully-automated body composition analyzer "InBody AI Scale"**
Consumer Electronic Show 2025
Developed height estimation model (accuracy 99.4%) and face recognition system of "InBody AI Scale".
Jan 2024 – Jan 2025
Las Vegas, USA
- **ECG-based cardiovascular diagnosis device using ML**
3rd Place (12 teams) — InBody Hackathon 2023 (\$1450)
Jul 2023 – Jul 2023
Seoul, South Korea
- **Autonomous mobile robot for executing missions**
3rd Place (20 teams) — AI Object Detection-Based Solar Tracking Robot Challenge (\$220)
Jul 2023 – Jul 2023
Seoul, South Korea
- **Dating app "Sidaeting"**
1,700 User Base
Implemented a matching algorithm that combinatorially optimizes collective happiness from users' ideal type data and developed the frontend for "Sidaeting", a dating service launched by UOSLIFE.
Mar 7 2022 – Jun 31 2022
Seoul, South Korea

SKILLS

- **Languages:** Korean, English
- **Technical Languages:** C, C++, Java, JavaScript, Julia, Python, SQL, HTML, CSS
- **Technical Frameworks:** ROS1-2 (Melodic, Noetic, Foxy, Humble, Jazzy), MuJoCo, PyTorch